SPEAKERDECK

Build instructions



A Spotify player and OctoPrint host with the size of one Denon SC-N10 speaker.

153 mm wide. 233 mm high. 196 mm deep.

Printed parts

	Part	File and Cura profile	Quantity	Size on the bed (mm)	Filament	Length
	Body	`stl/body.stl` Speakerdeck - Body 0.3	1	153 × 196 × 233	491 g	62.0 m
	Base plate	`stl/base.stl` Speakerdeck - Base 0.3	1	146 × 185 × 18	83 g	10.5 m
	Display frame	`stl/frame.stl` Speakerdeck - Display Frame 0.2	1	141 × 172 × 10	23 g	2.9 m
	Inky frame	`stl/inky_frame.stl` Speakerdeck - Inky Frame 0.1 slow	1	141 × 87 × 12	14 g	1.8 m
	Caps, 8 on one plate	`stl/cap.stl` Speakerdeck - Caps 0.2	1	67 × 31 × 7	6 g	0.8 m
	Inky pad	`stl/pad.stl` Speakerdeck - Pad 0.2	1	90 × 76 × 2	11 g	1.4 m
	Total				**628 g**	**79 m**

The filament values are estimates for PLA, 2.85 mm filament, 1.2 mm walls, and 15 % infill. The values for a part with a quantity include all pieces. A slicer gives the correct values for your settings.

- The STL files are in the `case/stl/` directory. Each file has the correct position for the print bed.
- Print the body upside down, with the top on the bed. Use supports for the display opening and the Inky cavity.
- Print the two frames in a dark color.
- The Cura profiles are for the LulzBot TAZ Workhorse with the 0.5 mm tool head and 2.85 mm PLA. Run `case/cura/make_profiles.py --install` to install them.
- The pad is rigid in PLA. For a soft pad, set `pad_squeeze = 0.5` and print it in TPU, or cut it from 2 mm foam.

Hardware

16	M3 heat-set insert, 4 mm long
12	M3 × 6 bolt
4	M3 × 6 countersunk bolt
4	M3 × 8 countersunk bolt
4	M3 × 16 bolt and M3 nut
4	Speaker binding post (2 red, 2 black)
1	DC panel jack, 5.5 × 2.1 mm
4	Rubber foot, self-adhesive

Electronics

1	Raspberry Pi 5 + adapter + HiFiBerry Amp2, as one stack
1	Inky wHAT with a 40-pin ribbon cable, 200 mm
1	Adafruit 7 inch HDMI display with touch
1	2.5 inch SATA drive with a USB adapter board
1	30 mm fan
1	Micro HDMI to HDMI cable, right-angle HDMI plug
1	USB-A to micro USB cable, right-angle micro USB plug
1	Short USB cable for the drive

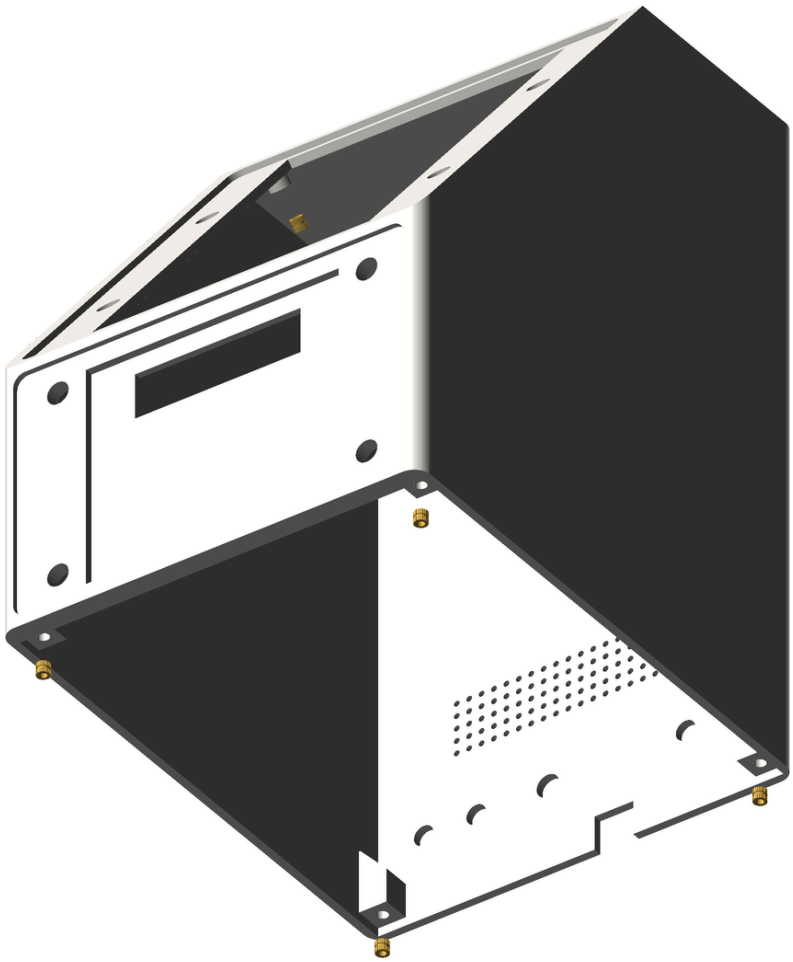
Tools: soldering iron with an insert tip, hex keys for M3, small flat screwdriver, calipers.

Measure first. Compare your Inky wHAT (thickness and header position) and the height of your Pi stack with the values in `speakerdeck.scad`. Correct the values, then run `build/make.sh`.

1 Inserts in the body

8 × M3 insert

Printed parts: `stl/body.stl`

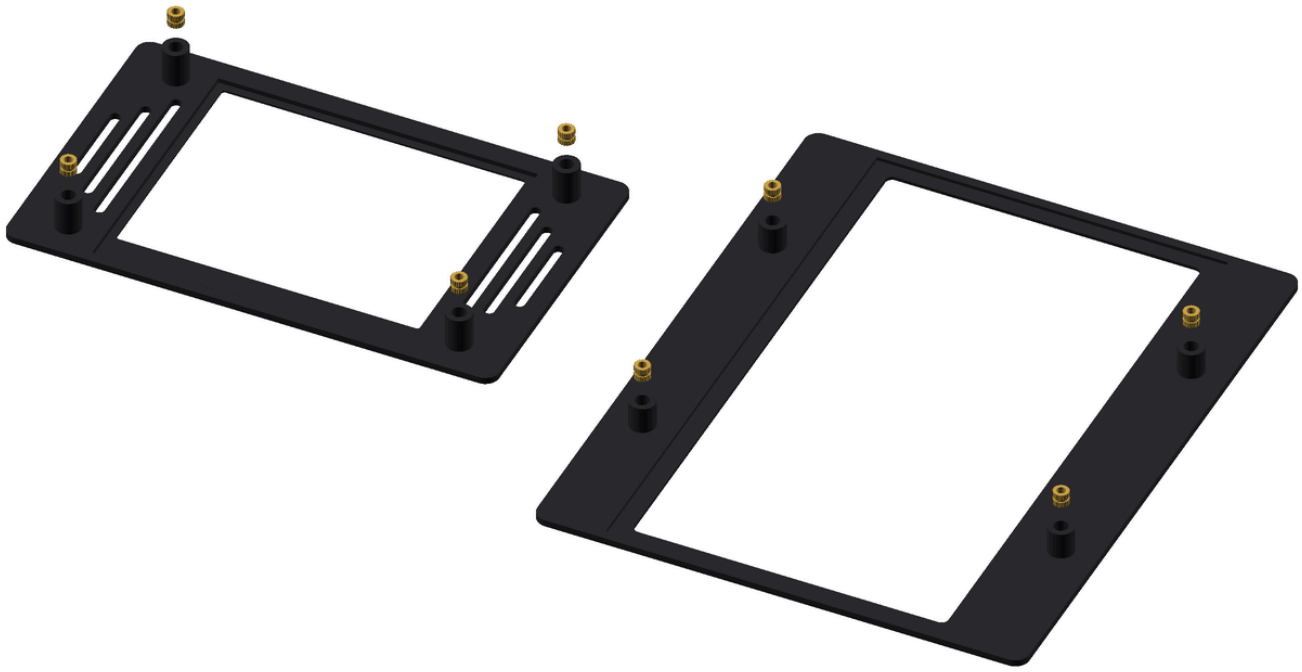


1. Put the body upside down. The picture shows the view from below.
2. Push 4 inserts into the 4 bosses around the display opening.
3. Push 4 inserts into the 4 corner posts.
4. Let the body become cool.

2 Inserts in the frames

8 × M3 insert

Printed parts: `stl/frame.stl`, `stl/inky_frame.stl`



1. Push 4 inserts into the 4 bosses on the rear face of the display frame.
2. Push 4 inserts into the 4 bosses on the rear face of the Inky frame.

3 Parts on the rear wall

4 × M3 × 16 + nut | **4 ×** Binding post | **1 ×** DC jack

Printed parts: `stl/body.stl`



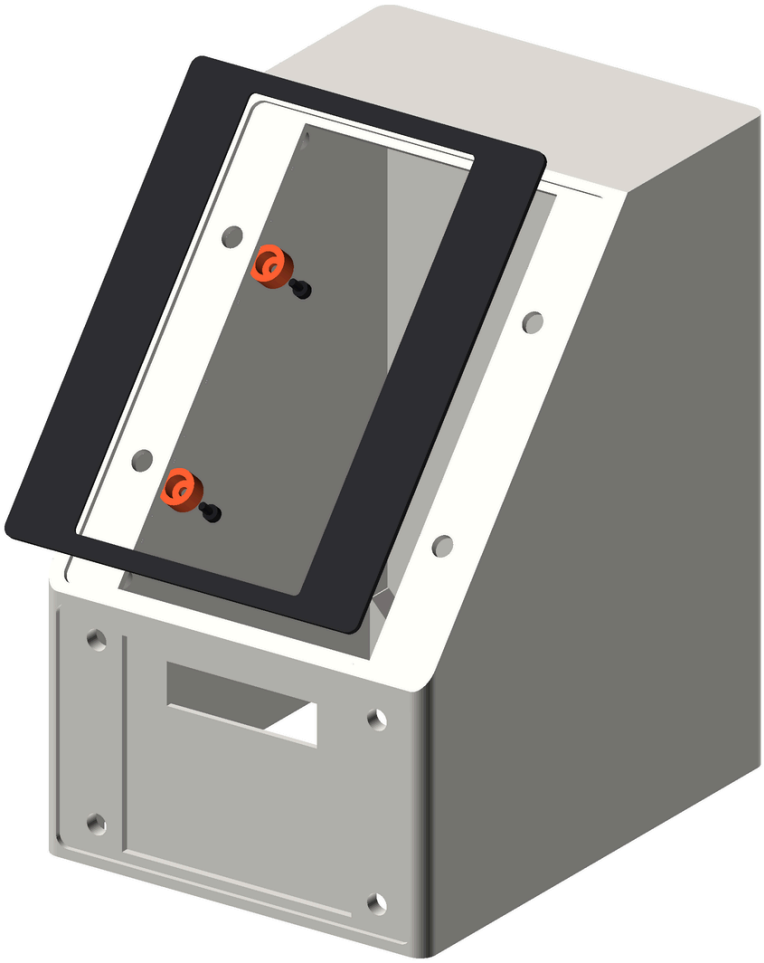
1. Attach the fan to the inner side, behind the round hole pattern. The label of the fan points to the wall, so the air goes out of the case.
2. Install the 4 binding posts. Put red on the inner holes and black on the outer holes.
3. Install the DC jack in the round hole near the center.

4 Display frame on the body

4 × Cap

4 × M3 × 6

Printed parts: `stl/body.stl`, `stl/frame.stl`, `stl/cap.stl`



1. Put the display frame into the indent in the angled front.
2. The 4 bosses of the frame go through the 4 holes of the body.
3. Put a cap on each boss from the inner side.
4. Install a bolt through each cap.

5 Display into the body

4 × M3 × 6

Printed parts: `stl/body.stl`



1. Connect the right-angle HDMI plug and the right-angle USB plug to the display.
2. Put the display into the body from below. The glass points to the front, and the sockets point down.
3. Put the LCD module into the opening. The board sits on the 4 bosses.
4. Install 4 bolts through the 4 holes of the display board.

The display glass is now 1 mm behind the front of the frame.

Caution. Tighten the bolts lightly. Too much force can bend the board.

6 Inky wHAT in the lower front

1 × Pad | **4 ×** Cap | **4 ×** M3 × 6

Printed parts: `stl/inky_frame.stl`, `stl/pad.stl`, `stl/cap.stl`



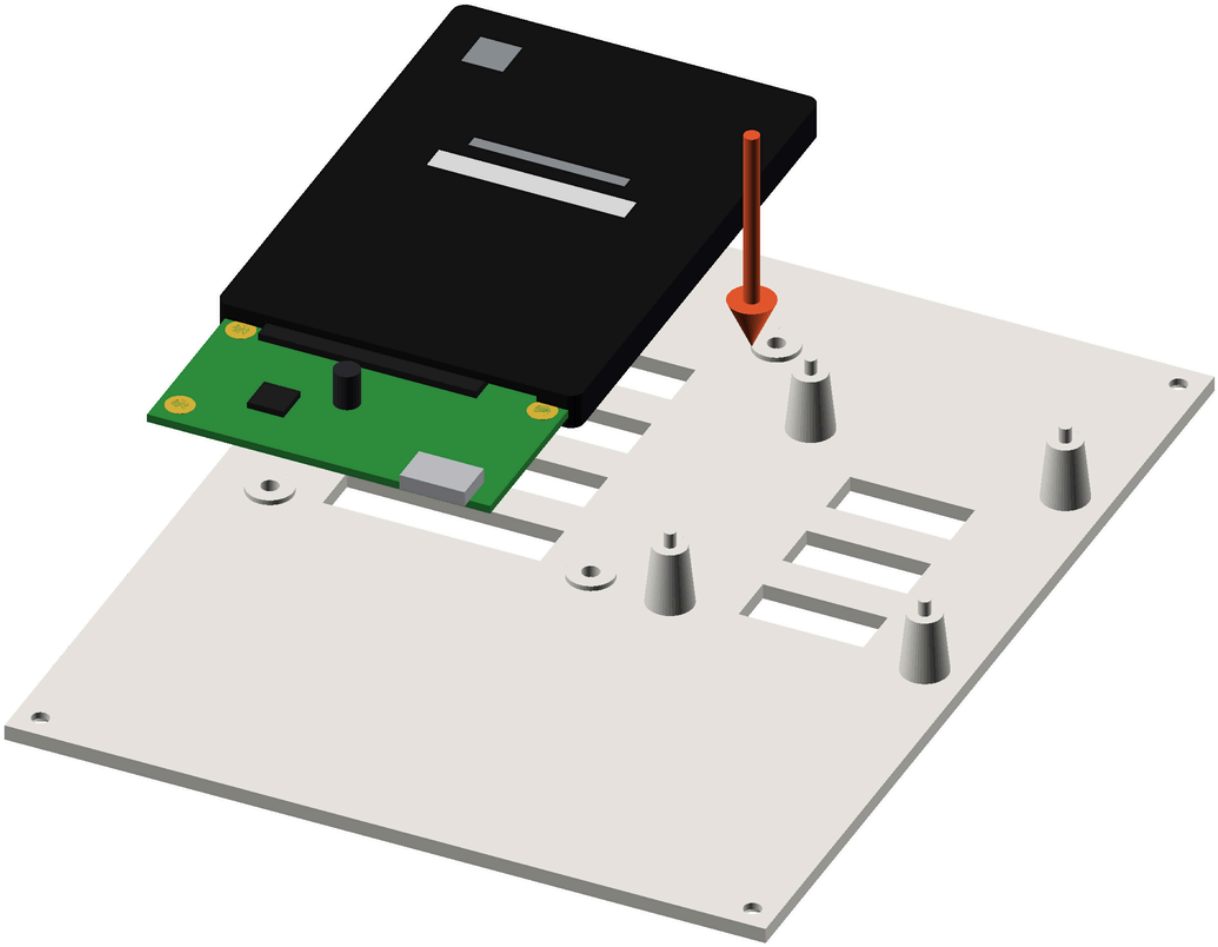
1. Put the pad into the cavity in the lower front.
2. Connect the ribbon cable to the header of the Inky wHAT.
3. Put the ribbon cable through the opening in the cavity.
4. Put the Inky wHAT on the pad. The wide border is at the bottom.
5. Put the Inky frame into the indent. The frame pushes the Inky wHAT on the pad.
6. Put a cap on each boss from the inner side, and install a bolt through each cap.

Caution. Do not push on the e-paper. The glass is thin.

7 Drive on the base plate

4 × M3 × 6 countersunk

Printed parts: `stl/base.stl`

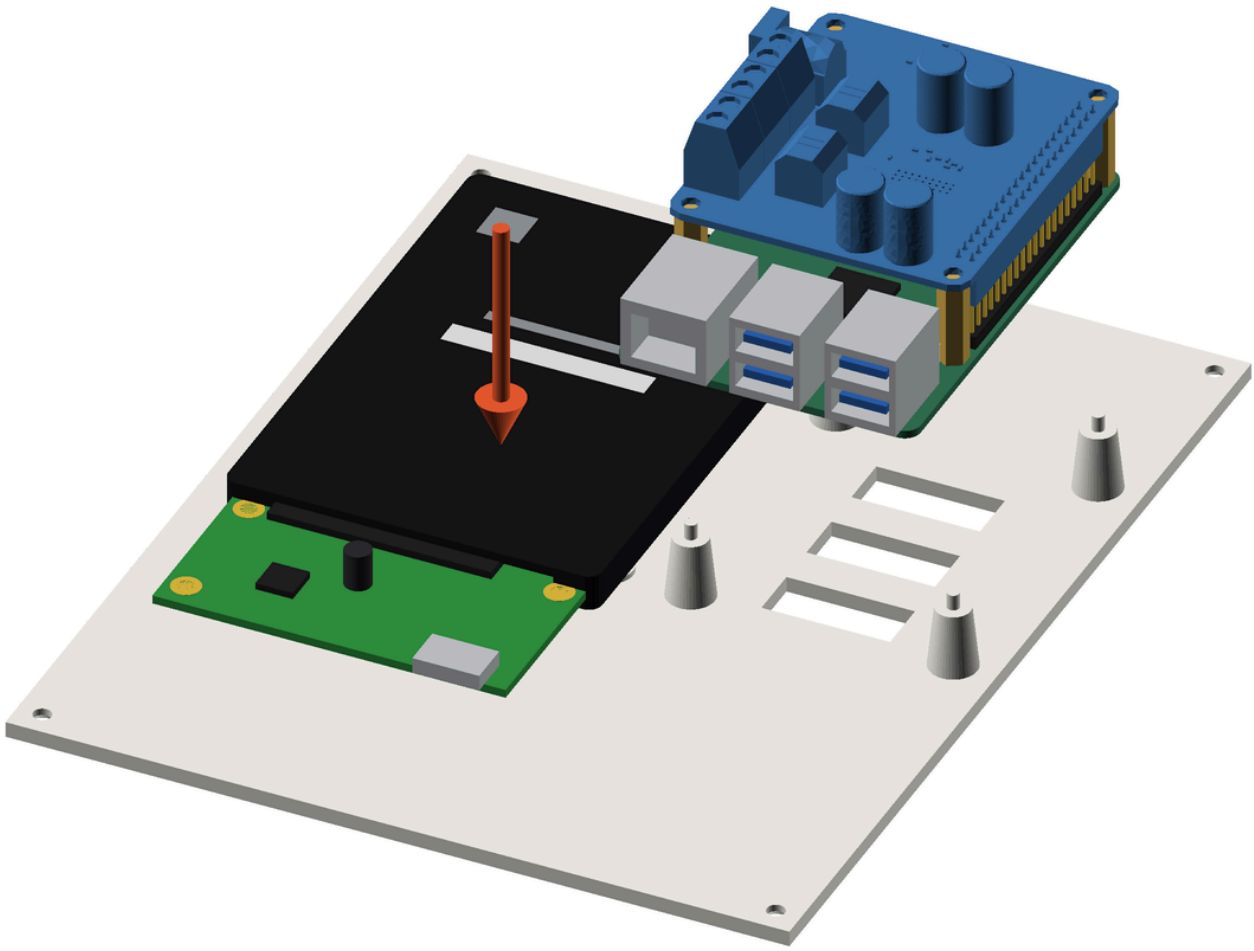


1. Put the drive on the 4 pads of the base plate. The label points down, and the adapter board points to the rear.
2. Install 4 countersunk bolts from below, through the base plate, into the 4 holes in the bottom of the drive.

Caution. Use M3 × 6 bolts or shorter. A longer bolt can damage the drive.

8 Pi stack on the base plate

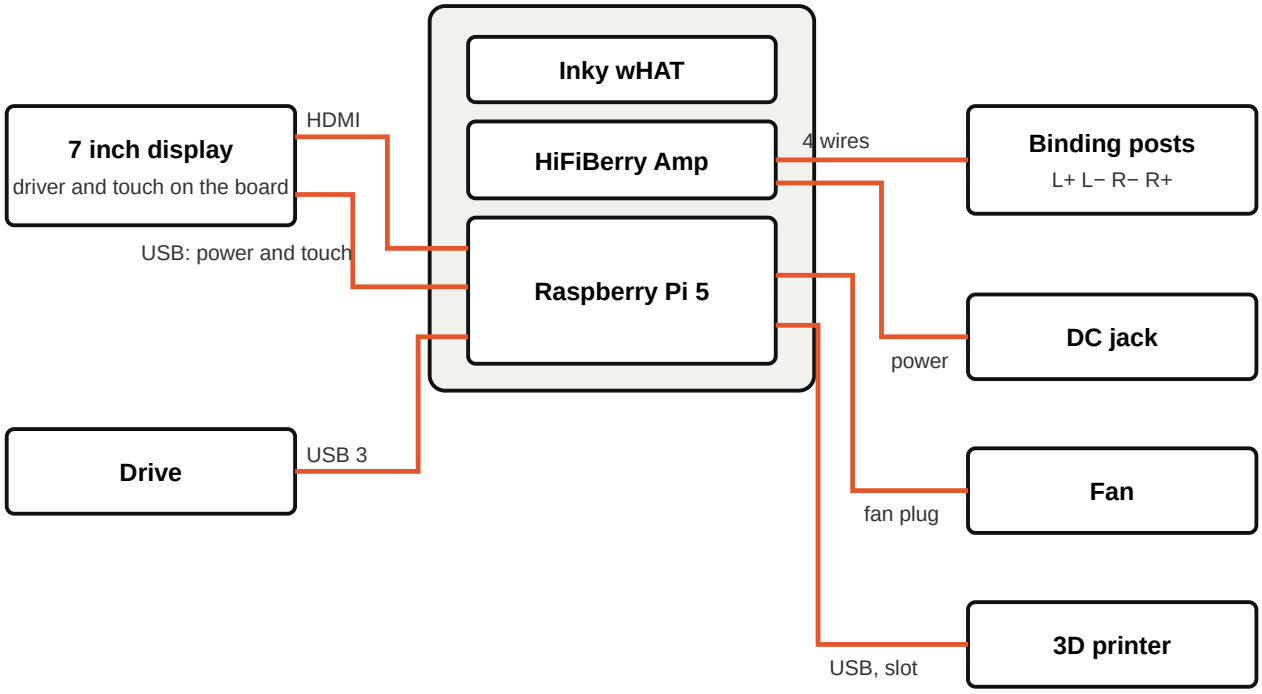
Printed parts: `stl/base.stl`



1. Put the Pi stack on the 4 pins of the base plate.
2. The USB ports point to the rear. The HDMI ports point to the drive.

The pins and the weight of the stack hold it in position. Keep the case upright.

9 Cables



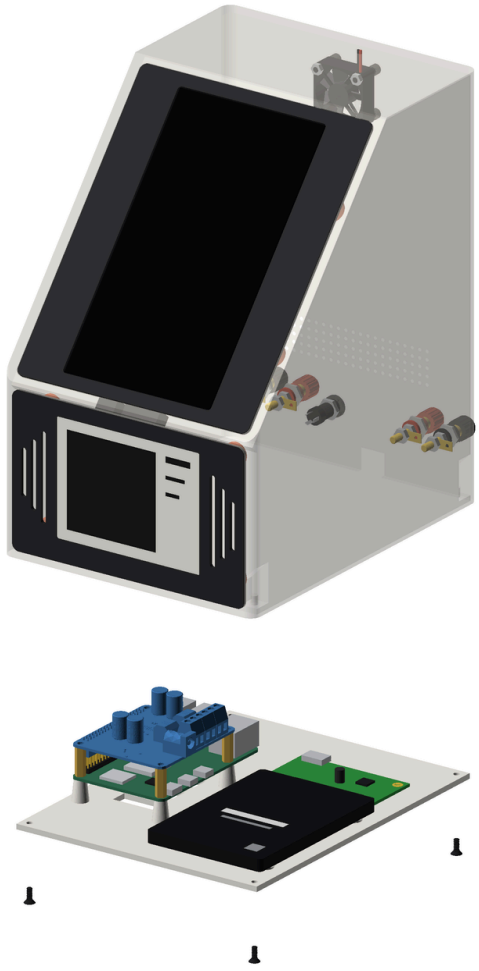
1. Put the body on its side, adjacent to the base plate.
2. Connect the HDMI cable of the display to a micro HDMI port of the Pi.
3. Connect the USB cable of the display to a USB port of the Pi.
4. Connect the drive to a blue USB port of the Pi.
5. Connect the ribbon cable of the Inky wHAT to the header of the stack.
6. Connect the binding posts to the speaker terminals of the Amp2. Keep the polarity correct.
7. Connect the DC jack to the power input of the Amp2.
8. Connect the fan to the fan connector of the Pi.

Warning. The Amp2 supplies power to the Pi. Do not connect a USB-C power supply at the same time.

10 Close the case

4 × M3 × 8 countersunk

Printed parts: `stl/body.stl`, `stl/base.stl`



1. Put the USB cable of the printer in the notch at the bottom of the rear wall.
2. Put the body on the base plate. Make sure that no cable is between the two parts.
3. Install 4 countersunk bolts from below, into the 4 corner posts.

11 Complete

4 × Rubber foot



1. Attach the 4 rubber feet to the bottom.
2. Put the unit between the speakers.
3. Connect the speaker cables to the binding posts.
4. Connect the 12 to 24 V power supply.